



IT-SIMPLERA INSTITUTE

INTERNSHIP WEEKLY REPORT

SUMMARY REPORT

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Role	Red Team Intern
Institute	IT-SIMPLERA Institute
Instructor	Talal Rehan
Week	Week 06
Submission Date	8 August 2026

This report presents the design of an Active Directory environment for TechSecure Solutions, a company with 300 employees, five departments, and two branch offices. The objective of this project is to create a secure and well-organized Active Directory infrastructure that supports user management, resource sharing, and centralized administration.

The report includes the design of the Active Directory structure, Organizational Units (OUs), user accounts, security groups, shared folder permissions, and recommended Group Policy settings. In addition, it explains how these configurations improve security by enforcing strong authentication, restricting unauthorized access, and reducing common cybersecurity risks. Overall, this assignment demonstrates the importance of proper Active Directory planning in maintaining a secure enterprise network.

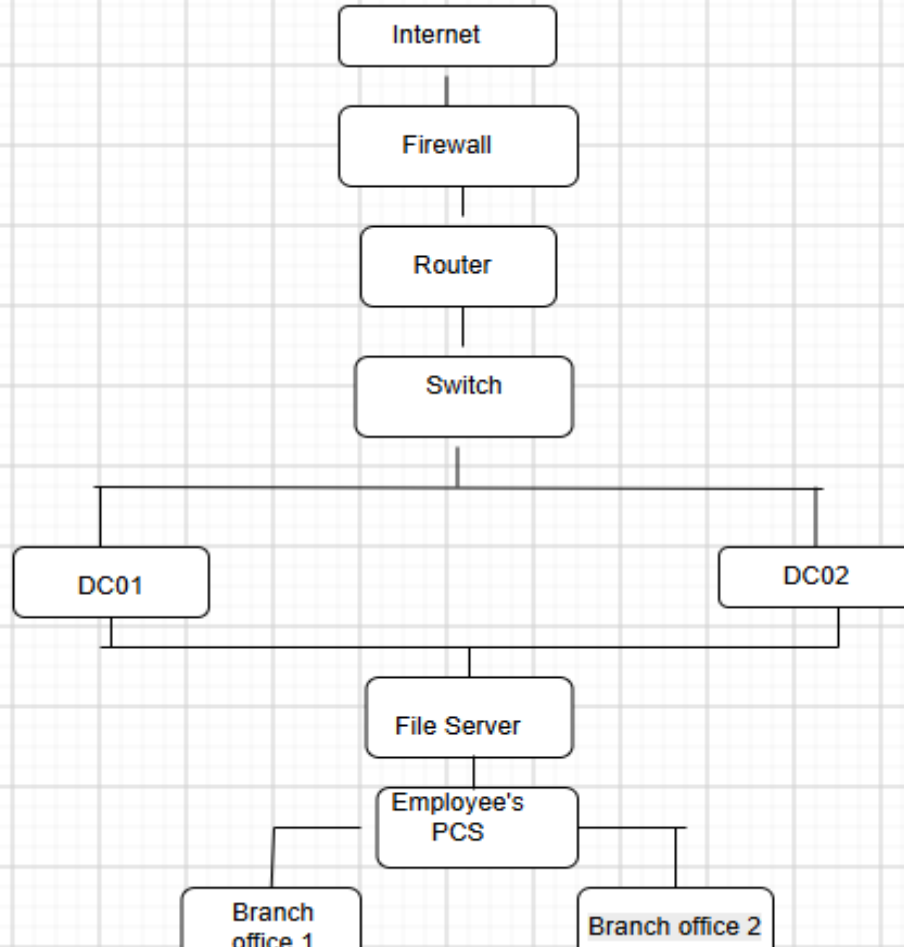
OBJECTIVE

The objective of this assignment is to design a secure Active Directory infrastructure for TechSecure Solutions using industry best practices. The project focuses on organizing users and departments, implementing appropriate security groups and permissions, planning Group Policy settings, and understanding how centralized administration improves security and management. It also aims to strengthen practical knowledge of enterprise network administration and Active Directory security concepts.

Environment

Component	Details
Operating System (Host)	Windows (Host Machine)
Virtualization Software	Oracle VirtualBox
Attacker VM	
Target VM	windows server 2022
Active Directory Domain	techsecure.local
Tools	Nmap, nslookup,dig,
Target Role	Domain Controll

TASK NO. 1



TechSecure.local

HR

IT

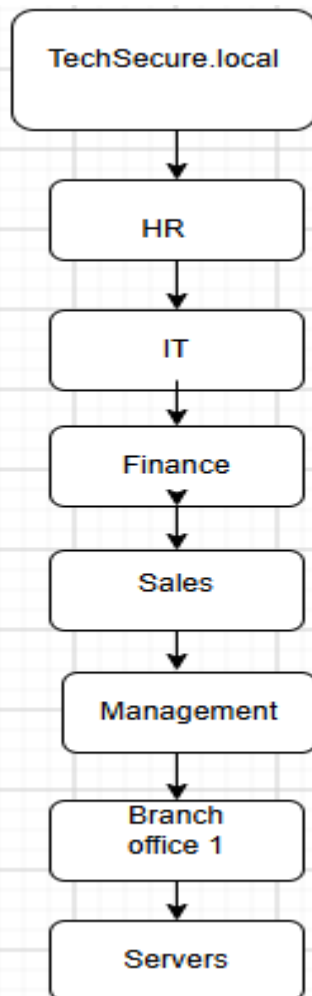
Finance

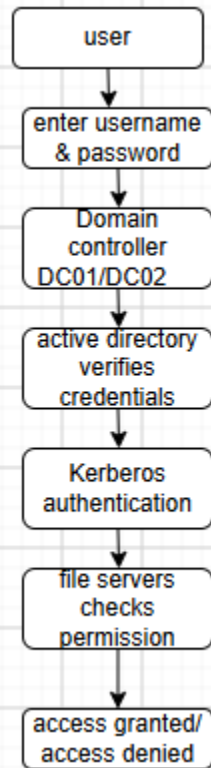
Sales

Management

Branch
office 1

Servers





TASK NO.2

Department	job title	User name	access rights	Full name	Group names
HR	HR Officer	akhan	HR Folder (Modify)	Ali Khan	HR_Group
HR	HR Assistant	sahemd	HR Folder (Modify)	Sara Ahmad	IT_Group
HR	Recruiter	araza	HR Folder (Modify)	Ahmad Raza	Finance_Group
HR	PayRoll officer	fnoor	HR Folder (Modify)	Fatima Noor	Sales_Group
HR	HR Manager	hali	HR Folder (control)	Hassan Ali	Management_Group

full name	epartment		roles	group	access rights
Bilal Hassan	IT	bhassan	system administrator	IT_Admins	IT_FOLDER (Full Control)
Ayesha Malik	IT	amalik	network admininstrator	IT_Admins	IT_FOLDER (Full Control)
Usman Tarik	IT	utarik	security analyst	IT_Admins	IT_FOLDER (Modify)
Hamza Iqbal	IT	hiqbal	help desk	Helpdesk	IT_FOLDER (Modify)
Zain Abbas	IT	zabbas	IT support engineer	IT_Admins	IT_FOLDER (Modify)

full name		department	roles	group	access rights
Muhammad Ali	mali	Finance	Finance Officer	Finance_Group	IT Folder (Modify)
Hira Khan	hkhan	Finance	Auditor	Finance_Group	IT Folder (Read)
Danish Malik	dmalik	Finance	Payroll executor	Finance_Group	IT Folder (Modify)
Sana Tarik	starik	Finance	Finance manager	Finance_Group	IT Folder (Full Control)
Imran Shan	ishah	Finance	accountant	Finance_Group	IT Folder (Modify)

full name		department	roles	group	access rights
Omar Siddiqi	ossidiqi	Sales	ales executive	Sales_Group	Sales_Folder(Modify)
Noor Fatima	nfatima	Sales	sales officer	Sales_Group	Sales_Folder(Modify)
Abdullah khan	akhan2	Sales	marketing executive	Sales_Group	Sales_Folder(Modify)
Maham Ali	mali2	Sales	Business developer officer	Sales_Group	Sales_Folder(Modify)
Saeed Ahmad	sahmad	Sales	sales manager	Sales_Group	Sales_Folder(Full Control)

Full name		department	roles	group	access right
Kamran Yousaf	kyousaf	Management	Chief executive officer	Management_Group	Management_folder(full control)
Areeba Malik	amalik2	Management	chief operating officer	Management_Group	Management_folder(full control
Fahad Khan	fkhan	Management	general manager	Management_Group	Management_folder(modify)
Iqra Ahmad	iahmad	Management	operational manager	Management_Group	Management_folder(modify)
Waqas Ali	wali	Management	project manager	Management_Group	Management_folder(modify)

Q1. Which account would attackers target first?

Answer

The Domain Administrator account would likely be the first target because it has the highest level of privileges in the Active Directory environment. If compromised, an attacker could control users, computers, security policies, and other critical resources.

Q2. Why?

Answer

The Domain Administrator has full administrative privileges across the domain. Compromising this account allows attackers to create new accounts, change permissions, access confidential data, install malicious software, and potentially take complete control of the network.

Folder	Access
HR	HR_Group
Finance	Finance_Group
Sales	Sales_Group
IT	IT_Group
Management	Management_Group
Public	All Employees(read only)

Q3. Which permissions violate the Least Privilege Principle?

Answer

Examples include:

- Sales users having Full Control over the HR folder.
- All employees having Full Control to shared folders.
- HR users having administrative privileges.
- Finance users accessing Management folders.
- Granting Full Control instead of Modify when it is not required.

These permissions provide users with more access than necessary, increasing the risk of accidental or malicious changes.

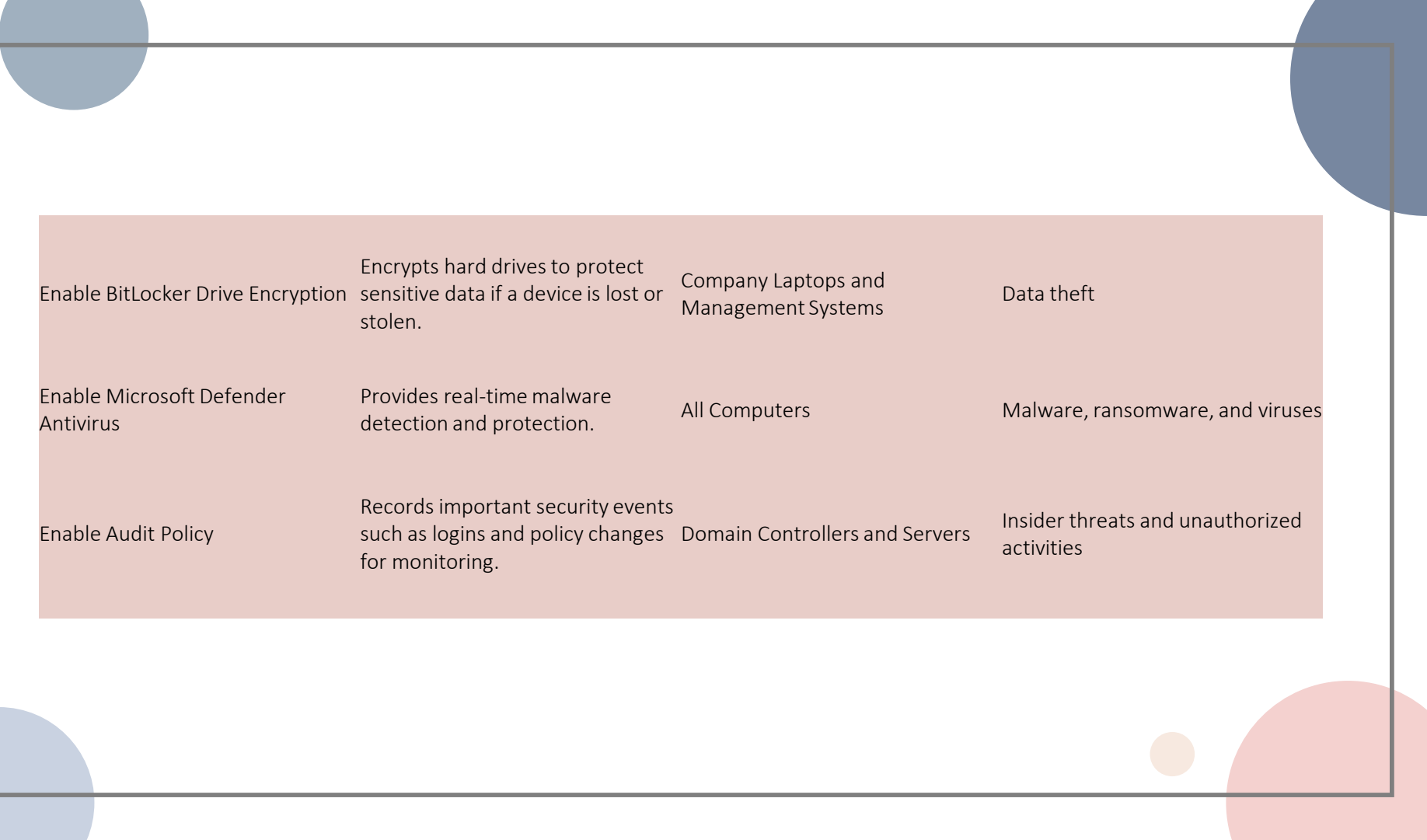
Q4. How would you fix those permissions?

Answer

The permissions should follow the Principle of Least Privilege by granting users only the access required for their job roles. Administrative privileges should be limited to IT administrators, department folders should only be accessible to their respective security groups, and Full Control should only be assigned to authorized administrators or department managers.

TASK NO.3

Group Policy	Why It Is Required	Users Affected	Cyberattack Prevented
Password Complexity	Enforces strong passwords that are difficult to guess.	All Domain Users	Password guessing and brute-force attacks
Account Lockout Policy	Locks an account after multiple failed login attempts.	All Domain Users	Brute-force and password spraying attacks
Disable USB Storage	Prevents unauthorized copying of company data to USB devices.	HR, Finance, Sales, Management	Data theft and malware infection
Disable Command Prompt (CMD)	Stops users from running unauthorized commands on company computers.	Standard Users	Unauthorized system changes and malware execution
Enable Windows Firewall	Protects systems by filtering incoming and outgoing network traffic.	All Computers	Network attacks and unauthorized access
Automatic Windows Updates	Ensures operating systems receive the latest security patches automatically.	All Computers	Exploitation of known vulnerabilities
Automatic Screen Lock	Locks inactive computers after a specified period of inactivity.	All Employees	Unauthorized physical access



Enable BitLocker Drive Encryption	Encrypts hard drives to protect sensitive data if a device is lost or stolen.	Company Laptops and Management Systems	Data theft
Enable Microsoft Defender Antivirus	Provides real-time malware detection and protection.	All Computers	Malware, ransomware, and viruses
Enable Audit Policy	Records important security events such as logins and policy changes for monitoring.	Domain Controllers and Servers	Insider threats and unauthorized activities

TASK NO.4

Task Details and Notifications

The screenshot shows the Windows Server Manager interface. The left-hand console tree is expanded to 'AD DS'. The main pane displays a list of servers under the heading 'SERVERS All servers | 1 total'. The table below shows the details of the single server listed.

Server Name	IPv4 Address	Manageability
WIN-6C3V4RPAKU8	10.0.2.15	Online - Perform...

Below the servers list, the 'EVENTS' section is visible, showing 'All events | 15 total'.

The 'Tools' menu is open, showing the following options:

- Active Directory Administrative Center
- Active Directory Domains and Trusts
- Active Directory Module for Windows PowerShell
- Active Directory Sites and Services
- Active Directory Users and Computers
- ADSI Edit
- Component Services
- Computer Management
- Defragment and Optimize Drives
- Disk Cleanup
- DNS
- Event Viewer
- Group Policy Management
- iSCSI Initiator
- Local Security Policy
- Microsoft Azure Services
- ODBC Data Sources (32-bit)
- ODBC Data Sources (64-bit)
- Performance Monitor
- Recovery Drive
- Registry Editor
- Resource Monitor
- Services

Task Details and Notifications

All Tasks |

Filter

Status

Dashboard

Local Server

All Servers

AD DS

DNS

File and Storage Services

Status

Notifications

Server Manager

← → ▾

AD DS

↺ ↻ | 🚩

Manage Tools View Help

SERVERS

All servers | 1 total

Filter 🔍 ⋮

Server Name	IPv4 Address	Manageability
WIN-6C3V4RPAKU8	10.0.2.15	Online - Perform

◀

▶

EVENTS

All events | 15 total

Filter 🔍 ⋮

Active Directory Administrative Center

Active Directory Domains and Trusts

Active Directory Module for Windows PowerShell

Active Directory Sites and Services

Active Directory Users and Computers

ADSI Edit

Component Services

Computer Management

Defragment and Optimize Drives

Disk Cleanup

DNS

Event Viewer

Group Policy Management

iSCSI Initiator

Local Security Policy

Microsoft Azure Services

ODBC Data Sources (32-bit)

ODBC Data Sources (64-bit)

Performance Monitor

Recovery Drive

Registry Editor

Resource Monitor

Services

Active Directory Users and Computers

File Action View Help



Active Directory Users and Computers [W

> Saved Queries

▼ techsecure.local

> Built-in

> Computers

> Domain Controllers

> ForeignSecurityPrincipals

> Managed Service Accounts

> Users

IT

Name

Type

Description

There are no items to show in this view.

Active Directory Users and Computers

File Action View Help



- Active Directory Users and Computers [W]
- > Saved Queries
- ▼ techsecure.local
 - > Built-in
 - > Computers
 - > Domain Controllers
 - > ForeignSecurityPrincipals
 - > Managed Service Accounts
 - > Users
 - IT

Name	Type	Description
Ali khan	User	
Ayesha noor	User	
hamza malik	User	
sara ahmad	User	
usman raza	User	

Active Directory Users and Computers

File Action View Help



- Active Directory Users and Computers [W]
- > Saved Queries
- ▼ techsecure.local
 - > Built-in
 - > Computers
 - > Domain Controllers
 - > ForeignSecurityPrincipals
 - > Managed Service Accounts
 - > Users
 - IT

Name	Type	Description
Ali khan	User	
Ayesha noor	User	
hamza malik	User	
IT_MANAGER	Security Group...	
IT_STAFF	Security Group...	
sara ahmad	User	
usman raza	User	

Active Directory Users and Computers

File Action View Help



Active Directory Users and Computers

> Saved Queries

▼ techsecure.local

> Built-in

> Computers

> Domain Controllers

> ForeignSecurityPrincipals

> Managed Service Accounts

> Users

IT

IT_STAFF Properties

General Members Member Of Managed By

Members:

Name	Active Directory Domain Services Folder
Ali khan	techsecure.local/IT
hamza malik	techsecure.local/IT
sara ahmad	techsecure.local/IT

Add...

Remove

OK

Cancel

Apply

Share View

> This PC > Desktop >

Name

Microsoft Edge

companyshare

5_X64

companyshare Properties

General Sharing Security Previous Versions Customize

Object name: C:\Users\Administrator\Desktop\companyshare

Group or user names:

IT_STAFF (TECHSECURE\IT_STAFF)
IT_MANAGER (TECHSECURE\IT_MANAGER)
Administrator
Administrators (TECHSECURE\Administrators)

To change permissions, click Edit.

Edit...

Permissions for SYSTEM

Allow

Deny

Full control

✓

Modify

✓

Read & execute

✓

List folder contents

✓

Read

✓

Write

✓

For special permissions or advanced settings, click Advanced.

Advanced

OK

Cancel

Apply

Active Directory Users and Computers

Action View Help



Active Directory Users and Computers [W

Saved Queries

chsecure.local

Builtin

Computers

Domain Controllers

ForeignSecurityPrincipals

Managed Service Accounts

Users

IT

Name	Type	Description
Ali khan	User	
Ayesha noor	User	
hamza malik	User	
IT_MANAGER	Security Group...	
IT_STAFF	Security Group...	
sara ahmad	User	

Active Directory Domain Services



The password for Ali khan has been changed.

OK

Active Directory Users and Computers

File Action View Help



Active Directory Users and Computers [W

> Saved Queries

▼ techsecure.local

> Built-in

> Computers

> Domain Controllers

> ForeignSecurityPrincipals

> Managed Service Accounts

> Users

IT

Name	Type	Description
------	------	-------------

Ali khan	User	
Ayesha noor	User	
hamza malik	User	
IT_MANAGER	Security Group...	
IT_STAFF	Security Group...	
sara ahmad	User	
usr		

Active Directory Domain Services



Object Ali khan has been disabled.

OK

TASK NO.5

```
(eman@kali)-[~]
$ nmap -sV -p 53,88,135,139,389,445,464,636,3268,3389 10.0.2.15
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-08 18:25 PKT
Nmap scan report for 10.0.2.15
Host is up (0.00014s latency).

PORT      STATE SERVICE      VERSION
53/tcp    closed domain
88/tcp    closed kerberos-sec
135/tcp   closed msrpc
139/tcp   closed netbios-ssn
389/tcp   closed ldap
445/tcp   closed microsoft-ds
464/tcp   closed kpasswd5
636/tcp   closed ldapssl
3268/tcp  closed globalcatLDAP
3389/tcp  closed ms-wbt-server

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 5.82 seconds

(eman@kali)-[~]
$ nslookup 10.0.2.15
** server can't find 15.2.0.10.in-addr.arpa: NXDOMAIN

(eman@kali)-[~]
```

Finding: The Windows Server at 10.0.2.15 was detected as an active host. However, the tested TCP ports associated with common Active Directory services, including DNS (53), Kerberos (88), LDAP (389), SMB (445), Global Catalog (3268), and RDP (3389), were reported as closed. This indicates that these services were not accessible from the Kali Linux system during the assessment.

Finding: The reverse DNS lookup for 10.0.2.15 returned NXDOMAIN, indicating that no PTR record for the target IP was available through the DNS server queried by Kali.

A terminal window on a Kali Linux system showing the execution of a reverse DNS lookup. The user enters the command 'dig -x 10.0.2.15'. The output shows the query details, including the question section for '15.2.0.10.in-addr.arpa. IN PTR' and the status 'NXDOMAIN'.

```
(eman@kali)-[~]  
$ dig -x 10.0.2.15  
  
; <<>> DiG 9.20.11-4+b1-Debian <<>> -x 10.0.2.15  
;; global options: +cmd  
;; Got answer:  
;; ->HEADER<- opcode: QUERY, status: NXDOMAIN, id: 36379  
;; flags: qr rd ra ad; QUERY: 1, ANSWER: 0, AUTHORITY: 0, ADDITIONAL: 0  
  
;; QUESTION SECTION:  
;15.2.0.10.in-addr.arpa.      IN      PTR  
  
;; Query time: 27 msec  
;; SERVER: 192.168.231.150#53(192.168.231.150) (UDP)  
;; WHEN: Sat Aug 08 18:29:48 PKT 2026  
;; MSG SIZE rcvd: 40  
  
(eman@kali)-[~]  
$
```

Finding: The dig -x reverse lookup also returned NXDOMAIN and no PTR record. This confirms that a reverse DNS name could not be resolved for the target IP through the configured DNS resolver.

